

AI Can Do Math Proofs: Why This is a Good Thing

By James squires

It's common knowledge that people are nervous about AI taking or replacing their jobs. Many academics, including some of my professors, have expressed similar concerns. And I don't fault people for this feeling. It seems the AI industry has endlessly hyped the ability of their AI models, even when it's outright silly to do so. This article isn't here to discuss the general limitations of AI though, but rather to discuss a specific application of AI in the context of math: the effects of AI doing math proofs.

AI Doing Math

Recently coding languages have been developed that allow us to code mathematical theorems and proofs. The coding languages check for mathematical correctness and throw an error if we make a mistake. This allows us to guarantee mathematical consistency as we go through a series of proofs. The same holds for AI though. We can have AI develop mathematical proofs in this language. Then we may choose to throw out all invalid proofs, thus giving us AI that can't hallucinate 'inconsistent math' (math with contradictions). This gives us a motif for *automatically proving mathematics* with AI.

The Role of a Mathematician

I would not fault someone for saying the mathematician's job is to prove theorems. I would argue this is wrong though. Let's take inspiration from the Ancient Greek etymology of mathematics: *μαθηματικά*, meaning "things pertaining to learning or study". We can find in

several Greek expositions (like Plato's Republic) that the belief in mathematics was that it strengthened understanding of the world's structure and essence. Mathematics was about determining what ought to be true through the use of abstraction. Notice the end-goal of mathematics wasn't to prove a theorem, but to rewrite one's perspective and understanding! This is what I claim the spirit of mathematics is: engineering one's mind toward a perspective of the world that recognizes underlying structures for what they have to be.

With this in mind, does AI threaten the mathematician? On a career level, I'd argue not. Even though AI can solve mathematical theorems, it's typically terrible at explaining them. Additionally, a solved theorem is not an understood theorem. The job of mathematicians is to communicate ideas and make ideas cogent. While AI is a good pedagogical tool, it is not a reliable communicator of liminal ideas. AI is not human, nor does it interact like one; AI makes probabilistic decisions based off of a token model. It does not have a model of the world nor human understanding (at least this is the common thought). Will it threaten a mathematician's job? Once again, I suspect the answer is no. For industry, the mathematician typically does computations, not logical proofs; due to the real world motivation behind most of these computational tasks, the industrial/applied mathematician, such as the actuarial or statistician, will not be supplanted by AI. What about the mathematics researcher? Well, what use is a proved theorem if no one in the world understands it? If we want to apply a theorem, we want to use a mathematical result to model the real world. AI cannot model the real world through the means of a human, such as senses, social interaction, etc, thus we still have the requisite need for modeling, understanding, and communication; all of which fall firmly on the hands of a human mathematician.

How AI Will Change the Math Landscape

AI will give us quicker access to solving theorems, allowing us to direct our efforts effectively. If we know a theorem is true, we now want to understand why and where this will lead us next.

Thus, we can think of automated theorem proving as an *oracle* that directs us toward our next goal of understanding. This is similar to how the computer allowed mathematicians to compute prime numbers or formulas; it gives us a layout of the landscape, except this time the landscape is that of logic.

[1] https://www.britannica.com/topic/Elements-by-Euclid?utm_source=chatgpt.com